Using agents to deal with high dimensions

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A former student's series

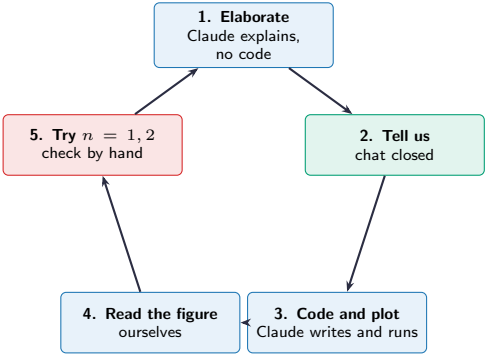


Chris Rowe wrote five short essays on what happens to simple geometry as the dimension grows.

Part	Title	The question	Today
1	Distance to the origin [29]	How far from the center is a random point of the unit ball?	Problem 1: the sphere
2	The cube inside the ball [31]	Does the unit cube fit inside the unit ball?	Problem 2: the cube
3 to 5	Distances between points, coin tosses, nearest neighbors [30, 28, 32]		

- Today we work through Parts 1 and 2 with Claude, in class.
- Each problem ends in a figure we check against an exact formula, and in its link to short-axis-sparse sketching operators (SASOs), a type of sparse random sketching operators [24, §2.4.1].
- Then we put the same problems through two custom Claude Code skills, an "answer key" and "office hours", and watch how the answer changes with the skill in use.

One loop for each problem



- In step 1 Claude is a tutor; in step 3 it is a programmer. Either can be wrong.
- Step 2 is the point of the lecture: if we cannot say it with the chat closed, we have not understood it yet.
- Step 4: each figure plots the simulated points and a curve computed from the exact formula. The curve is not fitted to the points, so if the points fall on the curve, the code and the math confirm each other. Read the axes before we read Claude's sentence about them.
- Step 5: at $n = 1$ (a segment) and $n = 2$ (a disk or a square) we can work out the answer by hand; compare it with the code's. In our rehearsal, $n = 1$ exposed a wrong formula that looked perfect for large n.

What a skill is

A **skill** is a folder of instructions, and optionally scripts or reference files, that teaches Claude how to do one kind of task. Until a task needs it, Claude sees only the skill's one-line description [1, 7].

- The folder holds a `SKILL.md` file: a short header (`name`, `description`) and plain-language instructions.
- Claude loads the body when a request matches the description, or when we call the skill by name; from then on it governs the rest of the chat [7].
- An open format: Claude Code, claude.ai, Codex and Gemini CLI read the same file [1, 11, 27, 16].

Where	How to add	How to call
Claude Code	A folder `.claude/skills/`*name*`/` holding `SKILL.md`, in the project or in our home folder	*/name*, or Claude picks it from the description
claude.ai, free plan too	Customize > Skills > + > Create skill > Upload a skill (a zip of the folder); needs code execution on	Claude picks it from the description

A skill marked `disable-model-invocation: true` runs in Claude Code only when typed. claude.ai rejects that key, so there the description must say when to use the skill [11].

Teach Claude who we are: a personal profile



Where the profile lives:

- Claude Code: `~/.claude/CLAUDE.md` for all our projects, `./CLAUDE.md` for one project [8].
- claude.ai: the personal instructions in Settings, which apply to every chat [10].
- Codex: `~/.codex/AGENTS.md` or `./AGENTS.md` [2].

What goes in it: who we are (Stat 157, Python + NumPy, what we have and have not taken); accuracy rules first; output format; hard limits ("never switch frameworks mid-answer").

The lazy way that works: let it interview us. In a *fresh* chat, paste:

```
Help me build a persistent context profile.  Ask ONE
question at a time about:  my role and courses; my
code stack; my accuracy standards; preferred output
format; what you should NEVER do; standing rules.
Validate each answer.  End with a complete Markdown
profile in my exact words, and save it as
~/.claude/CLAUDE.md; if that file already exists,
show me the changes and ask before overwriting it.
If you cannot write files, tell me where to paste
the profile instead.
```

Keep it fresh: monthly, "review my instructions against our recent chats and propose fixes." Keep it under a page; a bloated profile gets ignored [19].

Profile, project file, skill



Our profile (us, every project)

`~/.claude/CLAUDE.md`, what the interview produces, for example:

```
## How I want to work
- Explain the idea in two or three sentences
  before writing code.
- Show me the test before the implementation.
- Never change a test to make it pass; tell me
  why it fails.
- After making a figure, open it and tell me in
  one sentence what it shows.
```

A project file (this code)

`CLAUDE.md` in the project folder; other agents read `AGENTS.md` [8]. For example:

```
## Conventions
- The unit ball has radius 1; the unit cube is
  [-1/2, 1/2]^n; both centered at the origin.
- Vectorize with numpy; no Python loops over points.
- Keep each array at or below about 10^7 numbers.
- Print ASCII only, so scripts run on Windows too.
- Make the smallest change that does the task.
```

The profile says how *we* like to work, the project file says how *this code* works, and a skill says how to do *one kind of task*. Claude reads the first two at the start of every session; it loads a skill only when a task needs it.

A skill we use today: the plotting skill



```
---
name: sci-plot
description: House style for scientific matplotlib figures.
  Use whenever writing or changing code that draws a plot,
  histogram or figure.
---
1. One script per figure or topic; save PNG and PDF; no plt.show().
2. Simulation blue, exact formula red, references grey dashed;
   overlay the formula on every simulation that has one.
3. Label every axis in words; sample size in the legend.
4. Log axis when values span more than two decades. Zeros cannot
   sit on a log axis: drop them, draw a line at 1/(samples).
5. Panels that are compared share their x-axis.
6. Seed once at the top and pass rng to every sampler.
7. After saving, open the PNG and check it at the size it will be
   shown: no text below 7.5 pt, no text on a curve or a point,
   the formula on the simulation. Then tell me in one sentence
   what the figure shows.
8. Draw the figure at the size it will be shown; never scale it.
   For slides: about 2.5 x 2 in for half the width, 5 x 2.2 in
   for the full width. No text below 7.5 pt, legends included.
9. Keep text off the data: legends and labels in empty space or
   outside the axes. If a legend covers data, move it or shorten
   its labels and put the details in the caption.
```

- The description is written as a trigger ("use whenever..."), so Claude loads the skill by itself whenever it plots.
- Rule 4 will matter in Problem 1: a probability below $1/m$ cannot be estimated from m samples.
- Rules 8 and 9 came from preparing this deck: figures drawn large and shrunk onto a slide showed 5-point legends sitting on the data. Our figures now follow all nine rules.
- Rule 7 makes Claude look at its own figure. It does not make Claude's sentence about the figure right: step 4 of the loop is still ours.

Add-ons worth installing, by category



Name	Kind	Stars	What it does
Correctness signal			
`clangd-lsp` F	plugin	†	C/C++ diagnostics on every edit; find references
`pyright-lsp` F	plugin	†	same for Python (type and import errors)
`context7` C	MCP	∼62k	current, version-specific library docs [35]
Discipline, review, cleanup			
`superpowers` C	plugin	∼284k	spec → failing test → implement → debug [37]
`code-review` F	plugin	†	multi-agent PR review, confidence-scored
`code-simplifier` F	plugin	†	behavior-preserving cleanup of new code [5]
`security-guidance` F	plugin	†	flags injection, unsafe deserialization, leaked secrets
Tokens and cost			
`codeburn` O	CLI tool	∼11k	token and cost meter, 31 tools, no telemetry [15]
`caveman` C	skill	∼105k	terse output, about two-thirds fewer tokens [14]
Workflow			
`commit-commands` F	plugin	†	`/commit`, push, full PR in one bundle
`skill-creator` F	plugin	†	scaffold, evaluate, and audit our own skills [6]

F first-party · C community · O open-source tool; stars as of 9 Sep 2026, † first-party plugins share Anthropic's directory. All free. Install with `/plugin install name@claude-plugins-official`; a skill is a folder under `~/.claude/skills/` [6, 3]. The list and the commands are in the repository on the next slide. **The catch:** of 3,984 public skills Snyk audited, **36.8%** had a flaw, **76** carried malware [34]; the Verified badge does not cover bundled MCP servers [4].

Take the skills home



github.com/mmelnich/
stat157-agent-skills

- The profile-interview prompt, ready to copy.
- The plotting skill, and the answer-key and office-hours skills we use later today: for Claude Code, and as zips for claude.ai.
- The add-ons table with install commands, the install steps, and one link that downloads everything.

Problem 1: the sphere



PROBLEM 1 · PART 1 OF THE SERIES

Pick a point uniformly at random inside the unit ball in n dimensions.

How far from the center is it, on average?

How does that change as n grows, from $n = 1$ and $n = 2$ up to $n = 100$?

Problem 1: elaborate, then tell us



1.1, elaborate. Explain this problem to me without writing code: pick a point uniformly at random inside the unit ball in n dimensions. How far from the center is it, on average, and how does that change as n grows? Start with $n = 1$ (an interval) and $n = 2$ (a disk), then go to large n. Under 200 words; end with the one formula I should remember.

Tell us, with the chat closed

In the disk, is the average distance to the center $1/2$? What share of the 10-dimensional ball lies within half its radius of the center?

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Tell us, with the chat closed

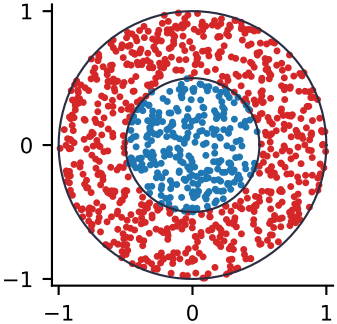
In the disk, is the average distance to the center $1/2$? What share of the 10-dimensional ball lies within half its radius of the center?

Let R be the distance to the center. The share of the ball within radius r is the volume ratio, and its derivative is the density of R:

$$\Pr(R \le r) = r^n, \qquad f(r) = \tfrac{d}{dr}\, r^n = n\, r^{n-1},$$

$$\mathbb{E}[R] = \int_0^1 r\, f(r)\, dr = \int_0^1 n\, r^n\, dr = \frac{n}{n+1}.$$

Disk ($n = 2$): $f(r) = 2r$, so $\mathbb{E}[R] = \int_0^1 r \cdot 2r\, dr = \frac{2}{3}$, not $\frac{1}{2}$; the guess $\frac{1}{2}$ is right only for $n = 1$. The 10-ball has $2^{-10} \approx 0.1\%$ of its volume within $\frac{1}{2}$ of the center.



$n = 2$, 1,000 uniform points: red beyond $\frac{1}{2}$ (72%; the exact share is 75%), blue within. Rings farther out hold more area, so they hold more points.

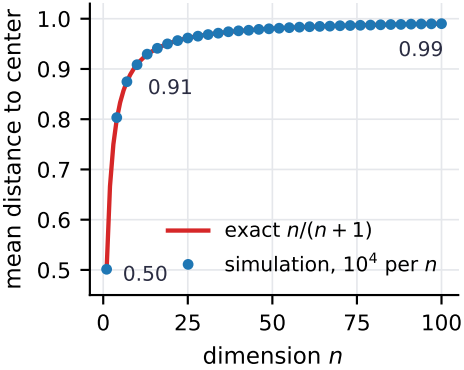
Problem 1: code and plot



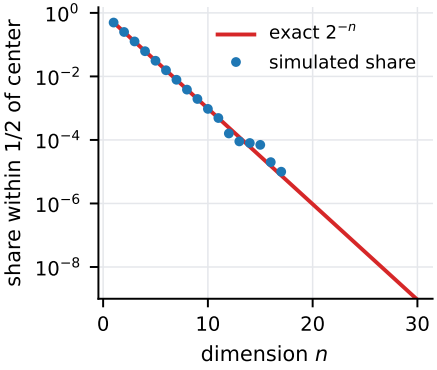
1.2, code and plot. Write a Python script that samples points uniformly in the unit n-ball and makes two figures: (a) the average distance to the center for $n = 1, \dots, 100$ (10,000 points each), with the exact curve $n/(n+1)$; (b) the share within $1/2$ of the center for $n = 1, \dots, 30$ (100,000 points each) on a log axis, with 2^{-n}. Run it, show me the figures, and tell me in one sentence what each shows.

Problem 1: code and plot

1.2, code and plot. Write a Python script that samples points uniformly in the unit n-ball and makes two figures: (a) the average distance to the center for $n = 1, \dots, 100$ (10,000 points each), with the exact curve $n/(n+1)$; (b) the share within $1/2$ of the center for $n = 1, \dots, 30$ (100,000 points each) on a log axis, with 2^{-n}. Run it, show me the figures, and tell me in one sentence what each shows.



(a) Mean distance 0.50 at $n = 1$ (the only dimension where the guess $\frac{1}{2}$ is right), 0.67 at $n = 2$, 0.91 at 10, 0.99 at 100: almost every point sits near the surface.

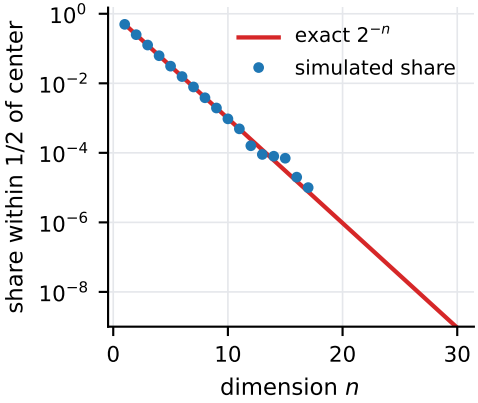


(b) The share within $\frac{1}{2}$ of the center follows 2^{-n} until the simulated points run out; the next slide asks why.

Problem 1: read the figure

Checkpoint question

We drew 100,000 points for each n. Why do the simulated points vanish from the log plot after $n = 17$?



Problem 1: read the figure

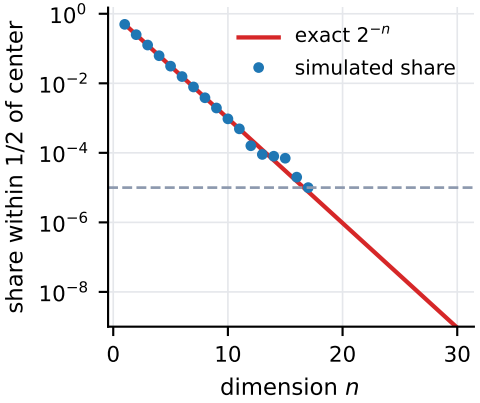
Checkpoint question

We drew 100,000 points for each n. Why do the simulated points vanish from the log plot after $n = 17$?

Each blue dot is a share: how many of the 100,000 points landed within $\frac{1}{2}$ of the center, divided by 100,000. The red line, 2^{-n}, is what that share should be on average.

n	16	17	18
points expected, $100{,}000 \times 2^{-n}$	1.5	0.76	0.38
points we got	2	1	0

Below one expected point we usually get 0, and a log axis cannot show 0. **Why:** with m samples the smallest nonzero share is $1/m$, one point, so a probability below about $1/m$ cannot be estimated. Here $1/m = 10^{-5}$ (the dashed line), and 2^{-n} falls below it at $n = 17$.



Problem 1: try $n = 1$ and $n = 2$

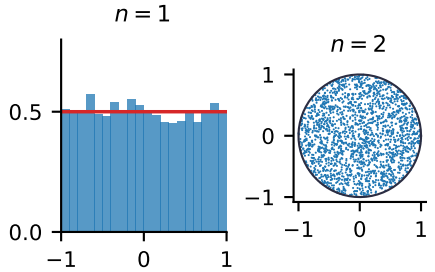


1.3, try $n = 1, 2$. Run your sampler at $n = 1$ and $n = 2$ and plot the points themselves: a histogram for $n = 1$, the disk for $n = 2$. Does the code still work? What should the mean distance be, and what do you get?

Problem 1: try $n = 1$ and $n = 2$

1.3, try $n = 1, 2$. Run your sampler at $n = 1$ and $n = 2$ and plot the points themselves: a histogram for $n = 1$, the disk for $n = 2$. Does the code still work? What should the mean distance be, and what do you get?

Work it out first. At $n = 1$ the ball is $[-1, 1]$ and the distance is uniform on $[0, 1]$: mean $\frac{1}{2}$. At $n = 2$: mean $\frac{2}{3}$.



2,000 points each, flat on $[-1, 1]$ and even over the disk. Mean distance 0.502 at $n = 1$ (exact $\frac{1}{2}$) and 0.669 at $n = 2$ (exact $\frac{2}{3}$).

Did the code break? Not in our rehearsal: in six runs (three each with Claude Sonnet 5.5 and Haiku 4.5 in Claude Code) the sampler ran at $n = 1$ and $n = 2$ and gave the right means.

The one $n = 1$ trap we know

Dividing by `norm(g, axis=1)` without `keepdims=True` crashes at $n = 2$, but at $n = 1$ numpy broadcasts the m points into an $m \times m$ array. At 2,000 points that is no error, just wrong numbers; at 100,000 it asks for 10^{10} entries and fails with a memory error.

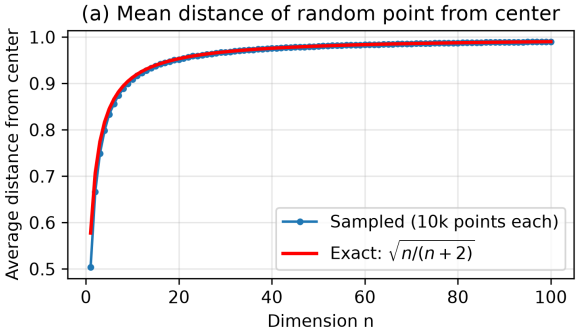
Problem 1: what $n = 1$ caught in our rehearsal



QUESTION

In our rehearsal the code ran fine at $n = 1$. What could $n = 1$ still catch?

Problem 1: what $n = 1$ caught in our rehearsal



1. **The agent got the wrong formula.** Asked to elaborate (step 1), Claude gave $\sqrt{n/(n+2)}$ as the mean distance; that is the root-mean-square distance, and the mean is $n/(n+1)$. In step 3 it plotted its own formula instead of the one we asked for.
2. **The curve looked good,** except at the smallest n. The gap between the two formulas is largest at $n = 1$ (0.577 against 0.500), 0.04 at $n = 2$, 0.01 at $n = 5$, and below 0.001 from $n = 21$ on. Claude called the match "tracking perfectly"; $n = 1$, where the mean is $\frac{1}{2}$ by hand, is where to look.

Claude Haiku 4.5, one rehearsal run; its own script, re-rendered on a smaller canvas so the text is readable.

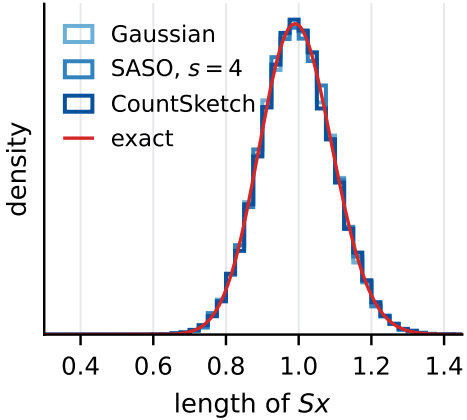
Problem 1 and sketching: a vector of length 1



QUESTION

Take a vector x of length 1 in $n = 1000$ dimensions. Multiply it by a random 50×1000 matrix S, a sketching operator: the result Sx, a vector with 50 entries, is the sketch of x. Is the length of Sx still about 1? Which S does it best: dense Gaussian entries, 4 nonzeros per column, or just 1?

Problem 1 and sketching: a vector of length 1



One vector x of length 1 in $n = 1000$ dimensions, multiplied by 20,000 random 50×1000 matrices S of each kind. Red: the exact law of $\|Sx\|$ for a Gaussian S, where $\|Sx\|^2 \sim \chi^2_k/k$.

Answer

Yes: $\|Sx\|$ stays near 1, about 1 ± 0.1; the spread is $1/\sqrt{2k} = 0.1$ for $k = 50$.

All three do equally well on this x, so the sparse ones win on cost: applying S takes s operations per entry of x instead of 50.

Why we care: with $\|Sx\| \approx \|x\|$, least squares and low-rank problems can be solved on the sketch, 50 numbers instead of 1000 [24].

Short-axis-sparse sketching operators (SASOs)

For a wide S the short-axis vectors are the columns. Each column has exactly s nonzeros, $\pm 1/\sqrt{s}$, in s distinct random rows [24, §2.4.1]. The family includes the sparse Johnson-Lindenstrauss transforms, CountSketch ($s = 1$) and OSNAPs [21, 25].

The same prompt without the plotting skill

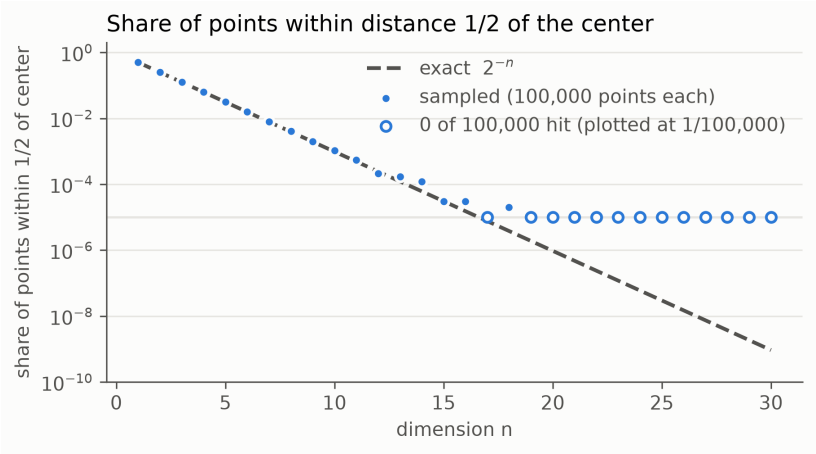


QUESTION

The same prompt 1.2, with no skill installed. How will Claude draw the dimensions where no sample landed within $\frac{1}{2}$?

The same prompt without the plotting skill

Prompt 1.2 in Claude Code (Claude Sonnet 5.5), one run in an empty folder. Figure (b), from Claude's own script, re-rendered on a smaller canvas so the text is readable:



The numbers are right, but dimensions with no sample inside $\frac{1}{2}$ are drawn as hollow markers at 10^{-5}: a row of points that reads like a floor in the data.

The same prompt with the plotting skill

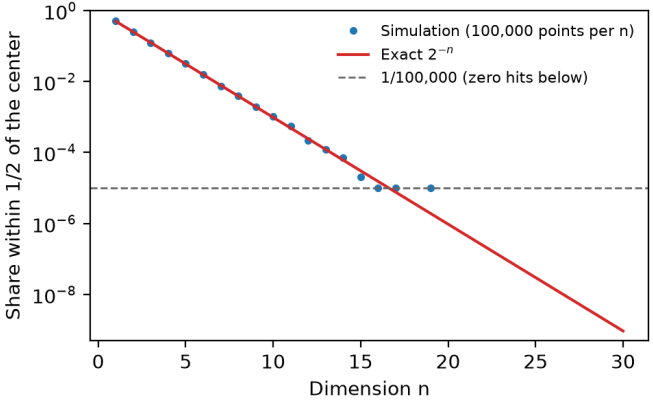


QUESTION

Now the same prompt with the plotting skill installed. What changes?

The same prompt with the plotting skill

The same prompt and model, one run in an empty folder that holds only the plotting skill (all nine rules). Figure (b), Claude's own, unmodified:



Zeros dropped and the floor a labelled line at one point in 10^5 (rule 4); formula red on blue; drawn at slide width with the legend in empty space (rules 8 and 9). Same model, prompt and numbers: the skill changed what the figure tells a reader.

Two tutors, one question



answer-key: the back of the book

```
1. First line: the final answer (a number,
   a formula or one sentence).
2. Then the complete worked solution, with
   every step shown. If the question involves
   code or a plot, give complete runnable code.
3. Do not ask the student anything: no "what
   do you think?", no hints instead of
   answers, no exercises.
4. Be correct, and state the assumptions you
   used.
```

office-hours: the professor's door

```
1. Start by asking what they already think or
   have tried. Do not explain anything yet.
2. One step per message, at most four
   sentences; then stop and wait.
3. Find the first wrong or missing step and
   ask about that step only.
4. When they are stuck, make the problem
   smaller: the smallest case, such as
   n = 1 or n = 2.
6. Never state the final answer, the full
   derivation or the complete code, even if
   asked. After three hints on one step,
   show that step only.
8. When they reach the answer, ask them to
   explain it back in two sentences.
```

Same model, same knowledge. Abridged here (office-hours items 5 and 7 omitted).

Live: the same problem through both skills



1. We ask Claude Code one question from today three times, each in a new session: no skill, `/answer-key`, `/office-hours`.
   - Why must the radius be $U^{1/n}$, and not U, to sample the ball uniformly?
   - Why does the cube stop fitting inside the ball, when its facet centers stay at $\frac{1}{2}$?
2. In office hours, the room plays the student: we type the room's answers, "I don't know" included.
3. Then we put the three answers side by side.

Compare

- Who wrote more, the student or Claude?
- Who did the computing?
- Could we rederive the answer tomorrow without the chat?
- Did either skill say something wrong, and would we have noticed?

Built-in versions of the second skill: in Claude Code, `/output-style learning` leaves `TODO(human)` for us to write [9]; ChatGPT has Study mode [26]; Gemini has Guided Learning [20].

Our rehearsal: one question, three tutors



QUESTION

One question to three tutors: no skill, answer key, office hours. Who writes the derivation in each?

Our rehearsal: one question, three tutors

Tutor: Claude Sonnet 5.5 in Claude Code with the two skills installed, a new session each time. The student in office hours was Claude Haiku 4.5 playing an undergraduate who has not seen the result: a simulation, not a student. Default means no skill.

Words written	Default	Answer key	Office hours
Sphere: tutor	326	591	474
Sphere: student	28	29	364
Cube: tutor	332	457	392
Cube: student	38	39	410
Messages	$1+1$	$1+1$	$6+6$
First to write the answer	tutor, line 1	tutor, line 1	student

Default and answer key both open with the result; the answer key adds runnable code and a table. Only in office hours does the student write the derivation.

Office hours, the sphere question (excerpts)

Tutor: ... predict this: if the points are truly uniform over the disk, what fraction should fall inside radius $1/2$? And what fraction with $R = U$?
Student: ... $1/4$ of the points? ... with $R = U$ ... half the points inside radius $1/2$? That seems like way more than $1/4$... is that the problem?
Tutor: Yes, that is the problem, and both of your computations are right. ... Now generalize your area argument.
Student: ... $P(R \le r) = r^2$ ... so $g(u) = \sqrt{u}$ ... that's $U^{1/2}$, which matches the pattern!
Tutor: ... right now it is a guess from one case. Now check it for general n.

Answers or guidance: what the research says



QUESTION

Does getting answers from AI help or hurt learning?

Answers or guidance: what the research says



High-school mathematics

−17%

on the later exam taken without AI, after practising with open GPT-4 (nearly 1,000 students). A tutor version that gave hints, not answers: no loss [12].

Developers, a new library

50% vs 67%

quiz score with and without AI help (52 people). Asking conceptual questions: 65% or more; handing the code to the AI: below 40% [33].

A course-wide assistant

−0.37 SD

in final grades: 0.37 standard deviations, about a third of the usual spread of grades between students (2,379 undergraduates). The GPT-4o assistant mostly answered directly (2026 working paper) [22].

In all three, getting the answer handed over went with learning less.

What to take home

- **The loop**: let Claude elaborate, say it back with the chat closed, then code and plot, read the figure ourselves, and try $n = 1$ and $n = 2$.
- **Small cases we can work out by hand** are cheap tests. In our rehearsal they caught a wrong formula, not a crash.
- **The sphere and the cube are in sketching**: a sketching operator's columns are points of the sphere; a SASO picks them among the cube's face centers; the sparse points of the cube are the ones that test it.
- **Set up once, choose on purpose**: a short profile, a project file, skills we have read. The answer key is fast; office hours is slower and leaves the derivation with us.

Skills and the profile prompt



`github.com/mmelnich/stat157-agent-skills`

Prompts for later

- 1.5 and 2.5: the sketching figures (backup slides).
- Parts 3 to 5 of the series [30, 28, 32].

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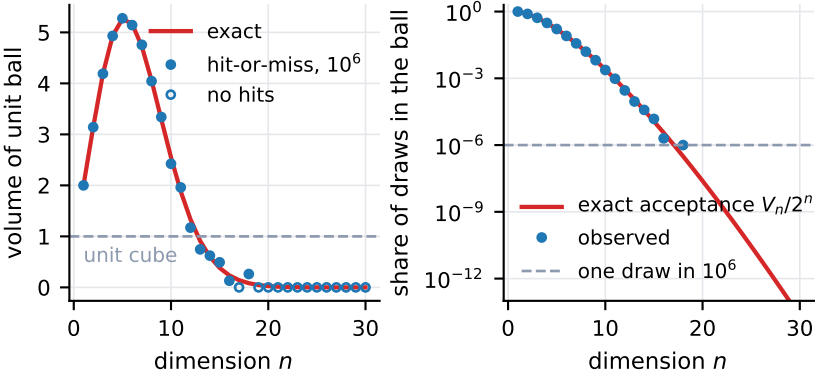
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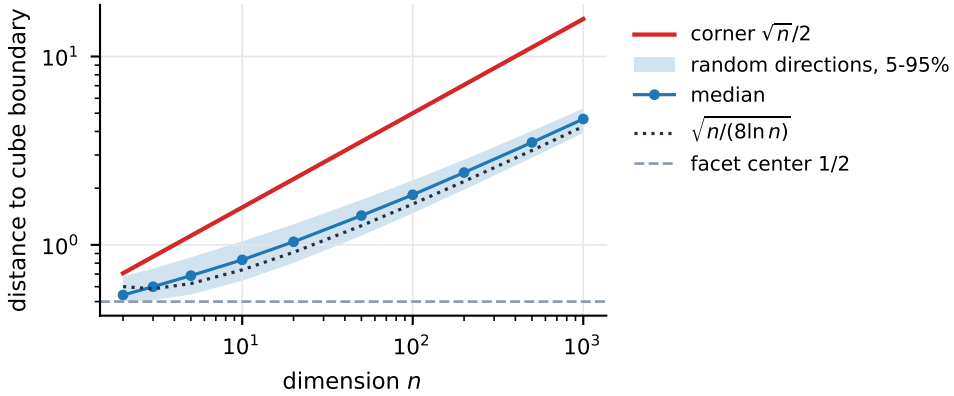
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Backup: the volume of the ball



The unit ball's volume peaks at $n = 5$ and drops below the unit cube's from $n = 13$. Hit-or-miss with 10^6 draws from $[-1, 1]^n$ finds no point of the ball after $n = 18$: rejection sampling cannot work there.

Backup: a convex body that looks spiky



Along a typical direction, the cube's boundary is about 1.8 from the center at $n = 100$ and 4.7 at $n = 1000$: far beyond $\frac{1}{2}$, far short of the corners. Both extremes are rare directions, which is how a convex body can look "spiky".

Backup: how to sample the ball uniformly



The standard sampler [23]

Direction: a Gaussian vector divided by its length, uniform on the sphere by rotational symmetry. Radius: $U^{1/n}$ with U uniform on $[0, 1]$, because the share inside radius r is r^n.

Two ways an agent gets it wrong

Radius U: the mean distance is $1/2$ in every dimension.
Rejection from the cube $[-1, 1]^n$: correct, but it keeps about 1 draw in 400 at $n = 10$ and 1 in 40 million at $n = 20$.

For questions about card 1.2 or the live question "why $U^{1/n}$ and not U?". In all six rehearsal runs Claude used the standard sampler.

Backup: extra prompts for the sketching figures



1.5, sketching a point of the sphere, extra. A sketching operator is a random $k \times n$ matrix S with $\|Sx\|$ close to $\|x\|$. Take one point x uniformly on the unit sphere in $n = 1000$ dimensions. With $k = 50$, draw 5,000 sketching operators of each kind and plot histograms of $\|Sx\|$: (a) Gaussian, entries $N(0, 1/k)$; (b) a short-axis-sparse sketching operator, where each column has $s = 4$ nonzeros equal to $\pm\frac{1}{2}$, in 4 distinct random rows; (c) CountSketch, the same with $s = 1$ and the nonzero equal to ± 1. Overlay the exact density of $\|Sx\|$ for (a): $\|Sx\|^2$ is chi-squared with k degrees of freedom, divided by k.

2.5, sketching a point of the cube, extra. Take $x = (1, 1, 0, \ldots, 0)/\sqrt{2}$ in $n = 1000$ dimensions: the center of a face of the cube with two nonzero coordinates, scaled to length 1. For $k = 10, 20, 50, 100$ and 200, estimate the probability that $\left|\|Sx\|^2 - 1\right| > \frac{1}{2}$ for CountSketch ($s = 1$) and for a short-axis-sparse sketching operator with $s = 4$, using 100,000 random draws of S each (only the two columns of S where x is nonzero matter). Plot both on log-log axes with the line $1/k$. Why is CountSketch's failure probability exactly $1/k$? Then try $x = (1, 0, \ldots, 0)$, a facet center: what happens, and why?

The figures on the two sketching slides used 20,000 random draws of S for 1.5 and 100,000 for 2.5.

Backup: the rest of the series

Part	Title	The one-line reason
3	The distance between points [30]	$\|x-y\|^2 = \|x\|^2 + \|y\|^2 - 2\langle x, y\rangle \approx 1 + 1 - 0$: random high-dimensional vectors are nearly orthogonal, so pairs sit near $\sqrt{2}$.
4	The probability connection [28]	Sums of n independent terms have relative spread $1/\sqrt{n}$ (coin tosses, the cube radius); the ball radius spreads like $1/n$ because $\Pr(\|x\| \le r) = r^n$.
5	The land of no neighbors [32]	A box holding 10% of uniform data needs $0.1^{1/n}$ of each axis: 79% at $n = 10$. Two numbers in Part 5 are slips (a power of ten; a pH scale of 15 instead of 14).

Backup: other agents



Skills are an open format, so the same `SKILL.md` works in these too; we rehearsed only Claude Code.

- Codex reads `AGENTS.md` and `.agents/skills/` [27].
- Gemini CLI, free with a personal Google account [17], reads `.agents/skills/` and `GEMINI.md` [18].
- Claude Code reads `.claude/skills/` only [7].

Backup: running the session



Proposed split (to adjust between us)

- One presenter: the loop and the live typing for Problem 1.
- The other: floor help during Problem 2, then the live run of the two skills. Both run the "tell us" rounds.

If the room runs late, cut in this order:

1. "Profile, project file, skill" and "Skills other people wrote": one minute each.
2. "Read the figure" for Problem 1: one minute.
3. The sketching slides: one sentence each.
4. The live run of the skills: show our rehearsal slide instead.

Fallbacks

- No code execution: paste the script into Python or Google Colab.
- No network: the slides carry every figure; pairs answer the "tell us" questions on paper.
- Agent loops on a fix: stop it after the third retry and ask a neighbor.